

Principles of Microeconomics — Week 3, Day 2

In-Class Problem Set: Repeated Games

This problem set accompanies the Day 2 session. Numbers differ from the lecture example. Solutions (instructor copy) are collected at the end; set `\solnfalse` in the preamble to produce a student version.

Both exercises use the same one-shot game — a Prisoner's Dilemma between two firms choosing to **Cooperate** (keep price high) or **Defect** (cut price). Row = firm 1; payoffs in \$m per period:

	Cooperate	Defect
Cooperate	(4, 4)	(0, 6)
Defect	(6, 0)	(2, 2)

Exercise 1 — A known ending

- (a) Find the Nash equilibrium of the one-shot game. Is Cooperate ever a best response?
- (b) The game is played exactly **three** times, and both firms know it. Using backward induction, work out what each firm does in period 3, then period 2, then period 1.
- (c) State the general lesson: for a game with a single one-shot Nash equilibrium, what happens over any *known, finite* number of repetitions, and why does the known ending matter?

Exercise 2 — No known ending, and the critical discount factor

Now the game is repeated **forever**, with a common discount factor δ between 0 and 1. Each firm uses a **grim-trigger** strategy: cooperate until someone defects, then play Defect forever. Recall that receiving π every period is worth $\pi/(1 - \delta)$.

- (a) Write the value of (i) cooperating forever, and (ii) defecting once and then being punished forever.
- (b) Find the critical discount factor δ^* above which cooperation is sustainable.
- (c) *Bigger temptation.* Suppose a successful defection now pays 8 instead of 6 (punishment unchanged at 2). Recompute δ^* . Does cooperation get easier or harder?
- (d) *Weaker punishment.* Back to a defection payoff of 6, but now the punishment (both-Defect) payoff is 3 instead of 2. Recompute δ^* and interpret.

Reconvene: state your δ^* for each case and, in one sentence, what makes tacit collusion *harder* to sustain.